

Regional Carbon Emission Analysis and Energy Efficiency Gap in Indian Data Centers

an uncertainty-aware assessment of seven major hubs

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Abstract

India's operational data-centre capacity has grown quickly enough that estimates of its electricity demand and carbon footprint are often built on installed megawatts rather than on how that capacity is actually used. This paper models facility electricity consumption and Scope 2 emissions for the seven largest Indian data-centre hubs as of H1 2026, using a transparent IT-load-factor approach rather than an assumption of continuous full-load operation. Under a central scenario of 65% average IT-load factor, a power usage effectiveness (PUE) of 1.45, and the Central Electricity Authority's FY2025–26 weighted-average grid emission factor of 0.675 t CO₂/MWh, the seven hubs are estimated to consume 14.49 TWh of facility electricity annually, producing 9.78 Mt CO₂/year on a location-based basis. A 50,000-draw Monte Carlo analysis shows that the IT-load factor, not server count or facility efficiency, is the dominant source of uncertainty in this estimate. The paper also separates location-based and market-based Scope 2 accounting, showing that naively discounting grid emissions by a renewable-procurement share double-counts renewable generation already embedded in the national grid average. The results are offered as a reproducible, assumption-transparent baseline for a fast-moving sector where public estimates have tended to vary by an order of magnitude.

Keywords: data centres; electricity demand; Scope 2 emissions; India; uncertainty analysis; carbon accounting

1. Introduction

India's data-centre sector has expanded rapidly over the past three years, driven by cloud migration, domestic data-localisation requirements and, more recently, demand tied to AI workloads. Coverage of the sector's electricity and carbon footprint has expanded alongside it, but a recurring weakness in both trade and early academic commentary is the treatment of installed capacity as if it were realised annual demand. A data centre rated at, say, 200 MW of IT load does not draw 200 MW around the clock; utilisation, redundancy and staged commissioning mean that actual draw is consistently well below nameplate capacity. Estimates that skip this step can overstate electricity consumption and downstream emissions by a wide margin, and once repeated in secondary sources, an inflated figure is difficult to correct.

This paper sets out a model that keeps that distinction explicit. It separates installed IT capacity, IT-load factor and facility-level power usage effectiveness, and it applies India's current, officially reported grid emission factor rather than assumed or dated city-level figures. It also treats

renewable procurement as a market-based accounting question rather than as a simple discount applied to location-based emissions, which matters increasingly as Indian operators announce larger renewable power purchase agreements. The aim is a baseline that other researchers can reproduce and update as better utilisation data becomes available, rather than a single point estimate presented without its assumptions.

2. Sector context

Cushman & Wakefield's September 2026 update puts India's operational data-centre capacity at close to 1.8 GW, with a development pipeline of roughly 3.9 GW, and identifies Mumbai as the country's largest single market by a wide margin [1, 2]. That growth needs to be read against the sector's actual share of national electricity demand, which remains modest: TERI estimated in September 2025 that Indian data centres accounted for under 1% of national electricity demand, with growth scenarios reaching as much as 6% by 2030 under an aggressive build-out [3]. Internationally, the IEA estimated global data-centre electricity consumption at around 415 TWh in 2024, about 1.5% of global electricity use, and its 2026 update projects roughly 485 TWh in 2025 rising to around 950 TWh by 2030 — a projection the IEA itself frames as uncertain given how AI deployment, hardware efficiency and grid constraints could each move the figure substantially [4].

This context matters for how the Indian numbers in this paper should be read. The sector is growing quickly in relative terms but starts from a small base, and the same caution the IEA applies to global 2030 projections applies equally to any single-point Indian capacity forecast.

3. Related work

Two strands of prior work inform the modelling choices made here. The first is the broader debate over whether data-centre electricity demand scales proportionally with computing workload. Masanet et al. showed that it does not: global data-centre electricity use grew only modestly between 2010 and 2018 despite a large increase in compute output, because server and facility efficiency gains largely offset workload growth [5]. That finding is a caution against extrapolating electricity demand directly from capacity or workload growth rates without an efficiency term. The second strand is bottom-up energy-use modelling of the kind used in Lawrence Berkeley National Laboratory's periodic United States data-centre energy reports, which build estimates from equipment counts, facility overheads and cooling assumptions and explicitly report ranges rather than single figures [6]. The model in this paper follows that second tradition at a smaller, hub-level scale appropriate to the data currently available for India.

On facility efficiency specifically, the Uptime Institute's 2026 global survey reports an average annual PUE across surveyed operators of 1.52, a capacity-weighted average of 1.36, and notes that newer large facilities routinely report figures at or below 1.3 [7]. There is not yet enough independently verified, publicly reported PUE data at the level of individual Indian cities to justify assigning different PUE values by hub; this paper therefore uses a single central PUE informed by that global evidence rather than fabricated regional precision.

4. Data and methodology

4.1 Capacity data

Operational IT capacity for the seven hubs is taken from the Cushman & Wakefield H1 2026 update [1, 2]:

Hub	Operational IT capacity
Mumbai	890 MW
Chennai	214 MW
Delhi-NCR	183 MW
Bengaluru	162 MW
Hyderabad	151 MW
Pune	133 MW
Kolkata	22 MW
Modelled total	1,755 MW

Table 1. Operational IT capacity by hub, H1 2026 [1, 2]. The seven-hub total (1,755 MW) sits about 34 MW, or 1.9%, below the headline top-seven total reported in the same update (1,789 MW); the gap is not rescaled and is instead carried through as capacity uncertainty in Section 6.

For each hub, annual IT-level electricity demand and total facility electricity demand are calculated as:

$$E_{IT} = C_{IT} \times LF \times 8760$$

$$E_{facility} = E_{IT} \times PUE$$

where C_{IT} is operational IT capacity (MW), LF is the average annual IT-load factor, 8760 converts MW to annual MWh, and PUE is the facility's power usage effectiveness. This two-step structure is the central methodological change from a capacity-only approach: it separates “how much IT equipment is installed” from “how much of that capacity is actually drawing power over a year”, and then adds the overhead of cooling, power distribution and other facility loads separately.

4.2 Central assumptions

Parameter	Central value	Basis
IT-load factor	0.65	Analytical assumption, varied in the Monte Carlo analysis
PUE	1.45	Informed by Uptime Institute 2026 survey evidence [7]
Grid emission factor	0.675 t CO ₂ /MWh	CEA FY2025–26 national weighted average [8]
Qualifying renewable procurement	40%	Market-based scenario variable only

Table 2. Central-scenario parameters.

The grid emission factor deserves particular note. The Central Electricity Authority (CEA) treats India as a single synchronised grid — unified since December 2013 — and reports one national

weighted-average factor rather than separate city-level figures [8]. For FY2025–26, that weighted average is 0.675 t CO₂/MWh, alongside a simple operating margin of 0.963, a build margin of 0.446 and a combined margin of 0.705 t CO₂/MWh. This paper uses the national weighted-average factor for location-based emissions and reserves the operating-margin figure for a specific market-based sensitivity described in Section 5. It does not assign different grid factors to individual cities, since CEA does not publish city-specific figures and doing so would introduce a false sense of regional precision.

As a plausibility check on this framing: even if Mumbai's 890 MW of IT capacity ran continuously at 100% utilisation with a PUE of 1.60, annual facility electricity would be about 12.47 TWh, and applying the current CEA average factor would put emissions at roughly 8.42 Mt CO₂/year for that hub alone — a fraction of the approximately 75 Mt CO₂/year that a full-load, dated-grid-factor assumption can produce. The gap illustrates why the load-factor and grid-factor choices, not just the underlying capacity figures, drive the final result.

5. Central-scenario results

Applying the central assumptions in Table 2 to the hub-level capacity data in Table 1 gives the following results.

Hub	Facility electricity (TWh/yr)	Location-based Scope 2 (Mt CO ₂ /yr)
Mumbai	7.348	4.960
Chennai	1.767	1.193
Delhi-NCR	1.511	1.020
Bengaluru	1.338	0.903
Hyderabad	1.247	0.842
Pune	1.098	0.741
Kolkata	0.182	0.123
Total	14.490	9.781

Table 3. Central-scenario facility electricity and location-based Scope 2 emissions by hub. These are scenario estimates, not operator-reported figures.

Figure 1 Central-scenario facility electricity and location-based Scope 2 emissions by hub, H1 2026 capacity, 65% IT-load factor, PUE 1.45, CEA FY2025–26 grid factor of 0.675 t CO₂/MWh.

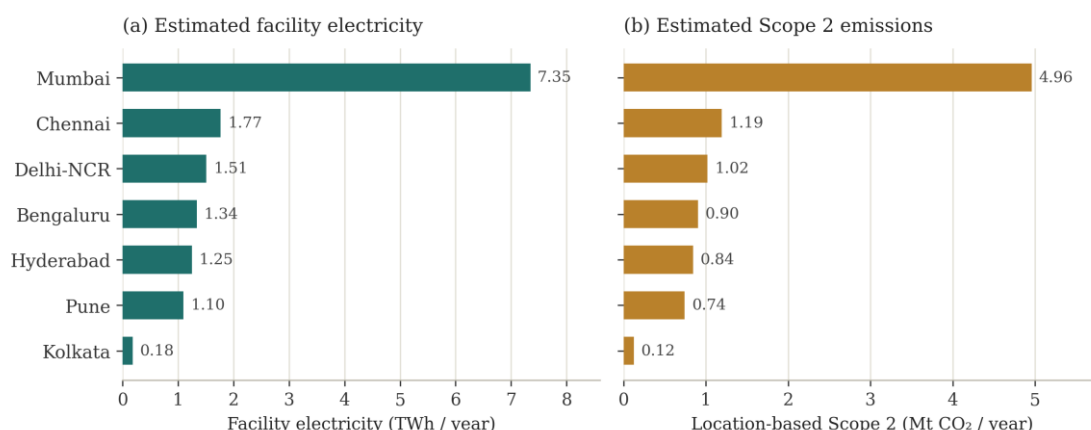


Figure 1. Central-scenario facility electricity and location-based Scope 2 emissions by hub.

Mumbai accounts for slightly over half of both modelled electricity demand and emissions, consistent with its position as by far the largest hub by installed capacity. Kolkata, at the other end, contributes under 2% of the total. This concentration matters for policy attention: interventions aimed at facility efficiency or renewable procurement in Mumbai alone would move the national total more than similar interventions spread evenly across all seven hubs.

6. Location-based versus market-based Scope 2

A common shortcut in earlier commentary on this sector is to take location-based emissions and multiply by (1 – renewable share) to arrive at a “net” figure. That shortcut is not consistent with GHG Protocol Scope 2 guidance, which treats location-based and market-based accounting as two distinct methods rather than one adjusting the other: location-based reflects the average emissions intensity of the grids on which consumption occurs, while market-based reflects the emissions associated with electricity that companies have specifically purchased through contractual instruments, with any unmatched consumption filled by a residual or grid-average factor [9].

$$Scope2_{LB} = E_{facility} \times EF_{grid}$$

$$Scope2_{MB} = E_{unmatched} \times EF_{unmatched} \text{ (+ matched renewable electricity, per contractual instrument)}$$

The distinction is not academic in this case. CEA's national weighted-average grid factor already incorporates the renewable generation that is on the Indian grid; discounting location-based emissions again by an assumed renewable-procurement percentage effectively counts part of that renewable benefit twice. For the central 40% qualifying-renewable market-based scenario, this paper instead reports two transparent bounds: a grid-average fallback of 5.87 Mt CO₂/year for unmatched consumption, and a more conservative sensitivity of 8.37 Mt CO₂/year using CEA's simple operating margin as the fallback factor rather than the blended national average. The operating-margin figure is used here only as a deliberately conservative sensitivity check, not as a claim about India's actual residual generation mix.

This is likely to become more consequential rather than less. The GHG Protocol's proposed Scope 2 revisions would tighten geographic deliverability and temporal matching requirements for renewable claims and have floated a fossil-only fallback factor where no defensible residual mix exists — changes that remain proposals rather than settled standards at the time of writing, but that point toward stricter market-based accounting in future [9]. Indian operators are, in the meantime, already moving toward higher contracted renewable shares: Nxtra announced in 2024 that additional wind and solar procurement would take renewable electricity to roughly 70% of its core data-centre energy mix, and followed in 2025 with a further 205,167 MWh of solar-wind hybrid supply [10]. That evidence is used here to bound the plausible range of renewable procurement scenarios, not to assume that every operator in the dataset has reached a 70% share.

7. Uncertainty and sensitivity analysis

Because IT-load factor, PUE, grid intensity and renewable procurement are all estimated rather than directly observed, the central-scenario figures in Table 3 are reported alongside a Monte Carlo analysis of 50,000 draws, varying capacity, IT-load factor, PUE, grid factor and qualifying renewable share independently within the ranges in Table 4. The analysis uses a fixed random seed (42) so that the results can be reproduced exactly.

Variable	Minimum	Central	Maximum
Capacity multiplier	0.95	1.00	1.05
IT-load factor	0.50	0.65	0.80
PUE	1.35	1.45	1.60
Grid factor (t CO ₂ /MWh)	0.64125	0.67500	0.70875
Qualifying renewable share	0%	40%	70%

Table 4. Monte Carlo input ranges. These are modelling distributions chosen to span plausible values, not measured statistical distributions.

Output	Mean	5th pct.	95th pct.
Facility electricity (TWh/yr)	14.66	12.20	17.19
Location-based Scope 2 (Mt CO ₂ /yr)	9.89	8.21	11.64
Market-based, grid fallback (Mt CO ₂ /yr)	6.26	3.84	9.02
Market-based, OM proxy (Mt CO ₂ /yr)	8.93	5.49	12.85

Table 5. Monte Carlo output distribution (n = 50,000).

Figure 2. Illustrative Monte Carlo distribution of aggregate location-based Scope 2 emissions across seven hubs (50,000 draws; capacity, IT-load factor, PUE and grid intensity varied independently; seed = 42).

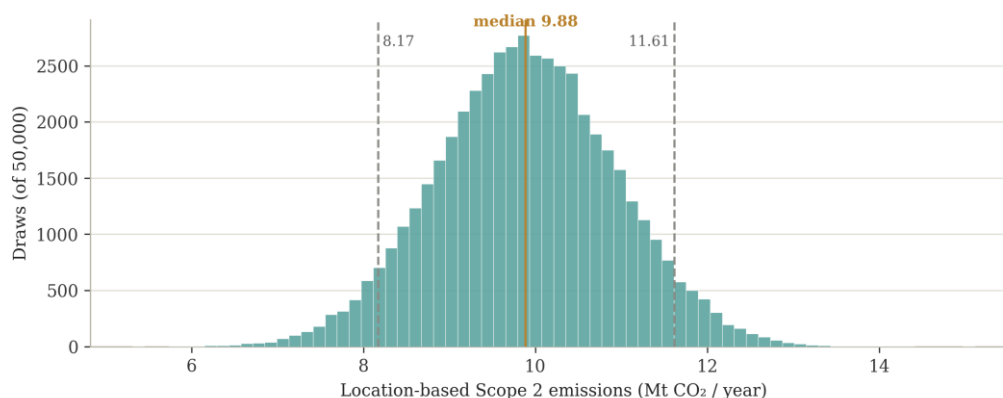


Figure 2. Illustrative Monte Carlo distribution of aggregate location-based Scope 2 emissions across the seven hubs.

Rank-correlation sensitivity analysis indicates that IT-load factor is by a wide margin the dominant source of uncertainty in location-based emissions ($\rho \approx 0.90$), with PUE a distant second ($\rho \approx 0.32$). For the market-based calculation, qualifying renewable procurement share dominates instead ($\rho \approx -0.91$). The practical implication is that better utilisation data — actual IT-load factors reported by operators or inferred from metered consumption — would narrow these estimates considerably more than further refinement of facility-level PUE assumptions, which is where most publicly available operational disclosure currently concentrates.

8. Discussion

The scale of the correction between a full-load, capacity-only estimate and the load-factor-adjusted estimate presented here — roughly an order of magnitude at the hub level — is large enough to change how the sector's footprint should be discussed publicly. Reporting installed megawatts as if they were annual electricity draw is an easy mistake to make because capacity figures are the most readily available public data point, but it systematically overstates both electricity demand and emissions, and errors of that size are difficult to walk back once a headline figure has circulated. Separating capacity, utilisation and facility overhead, and applying a current, officially sourced grid factor rather than an assumed or outdated one, produces a materially different and more defensible picture.

The distinction between location-based and market-based accounting matters for a second reason: as Indian operators publicise increasingly ambitious renewable procurement targets, there is a risk that market-based claims are read informally as reductions in location-based, grid-level impact. They are not the same claim, and conflating them understates the residual pressure that data-centre growth places on the shared Indian grid even where individual operators have contracted substantial renewable supply.

9. Limitations

Several limitations should be read alongside the results. Capacity figures for individual hubs come from a single commercial real-estate source and carry a disclosed 1.9% discrepancy against that

source's own headline total, which is treated as capacity uncertainty in the Monte Carlo analysis rather than resolved by rescaling. IT-load factor, PUE and renewable-share ranges are modelling assumptions informed by industry survey evidence rather than metered data for the specific facilities in scope, and they should be treated as such. The analysis also considers only the seven largest publicly reported hubs; smaller and emerging markets are excluded, so the results understate India's total data-centre footprint even as they aim to be accurate about the hubs they do cover. Finally, the market-based operating-margin sensitivity is explicitly a conservative proxy rather than a measured residual-mix factor, since India does not yet publish one.

10. Conclusion

Under a central, transparently stated scenario, India's seven largest data-centre hubs are estimated to consume 14.49 TWh of facility electricity and produce 9.78 Mt CO₂/year of location-based Scope 2 emissions as of H1 2026. That estimate is roughly an order of magnitude below what a full-load, capacity-only calculation would suggest, which illustrates how much the choice of methodology — not just the choice of input data — shapes public understanding of the sector's footprint.

Uncertainty analysis further shows that IT-load factor, not facility efficiency or grid intensity, is the single largest driver of that uncertainty, pointing to where future data collection would be most valuable. As Indian operators continue to expand both capacity and renewable procurement, keeping location-based and market-based accounting separate — rather than collapsing one into the other — will matter increasingly for how progress against emissions targets is judged.

Data availability: the hub-level dataset, scenario assumptions, Monte Carlo notebook (fixed seed, reproducible) and sensitivity outputs underlying this paper are available from the corresponding author on request.

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